

Experiment No : 02

Experiment Name: Study of half-wave and full-wave rectifier using diodes

Objectives:

- To implement the half-wave and full-wave rectifier using a p-n junction diode
- To observe the output waveforms of the half-wave and full-wave rectifier

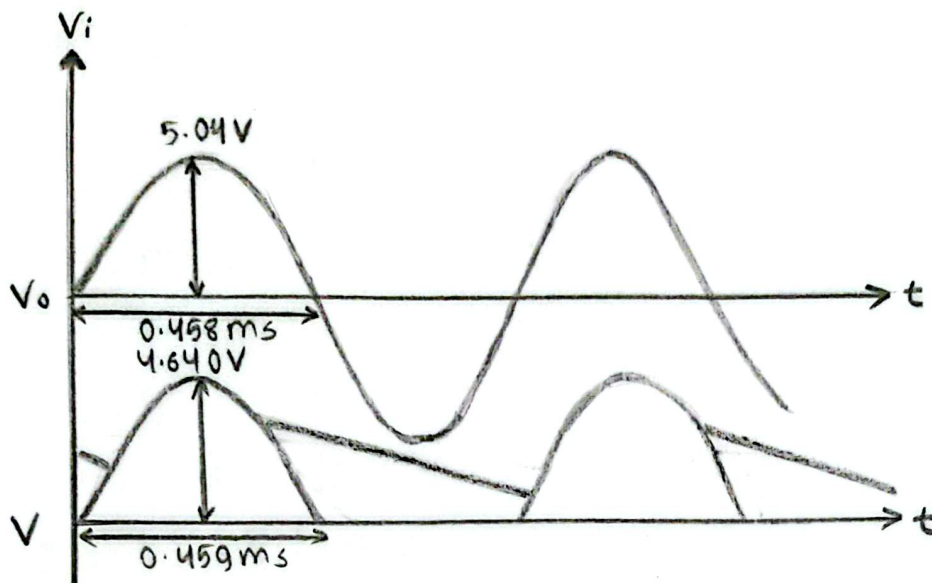
Theory:

A rectifier converts an AC signal into a DC signal. According to the diode's characteristics curve, conduction occurs exclusively under forward-bias conditions, while reverse bias effectively prevents current flow. Consequently, when an AC voltage is applied across the diode, it transmits only one half of the input cycle, either the positive or negative half, depending on the diode's orientation in the circuit. During forward bias, the diode conducts, allowing that half-cycle to pass, while the other half is blocked, so the load receives a pulsating DC signal.

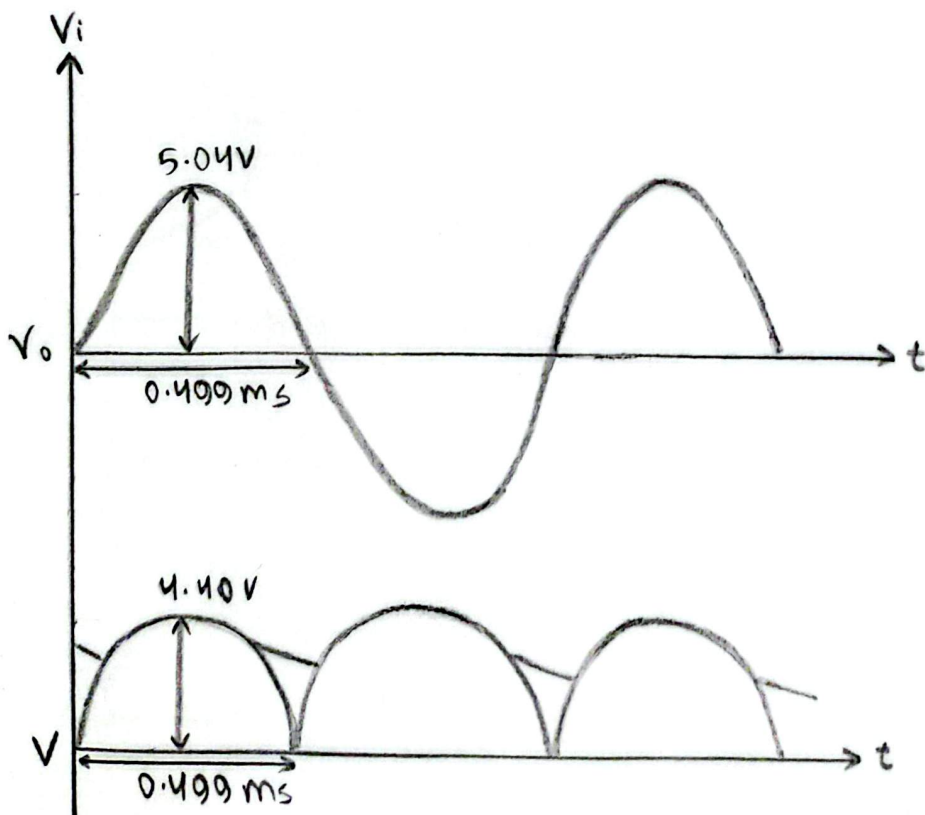
Diode rectifier can be categorized in two types—

- i. Half-Wave rectifier
- ii. Full-Wave rectifier

Half-Wave Rectifier:



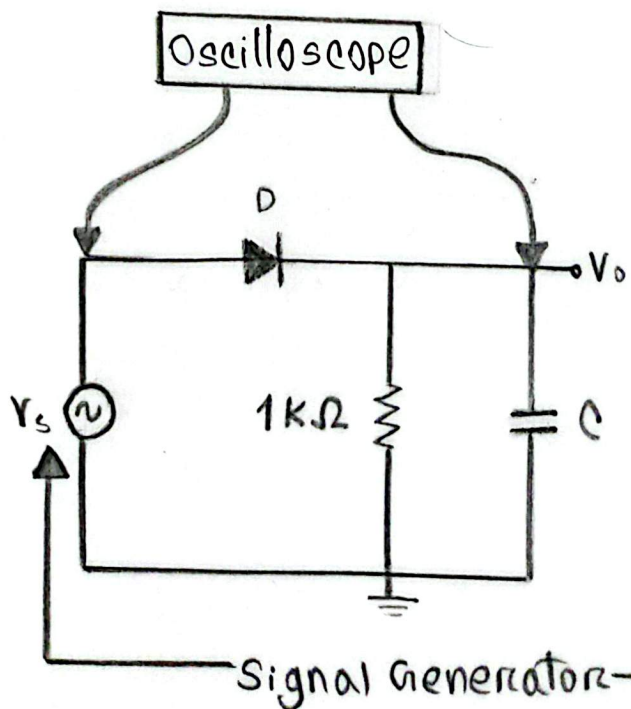
Full-Wave Rectifier:



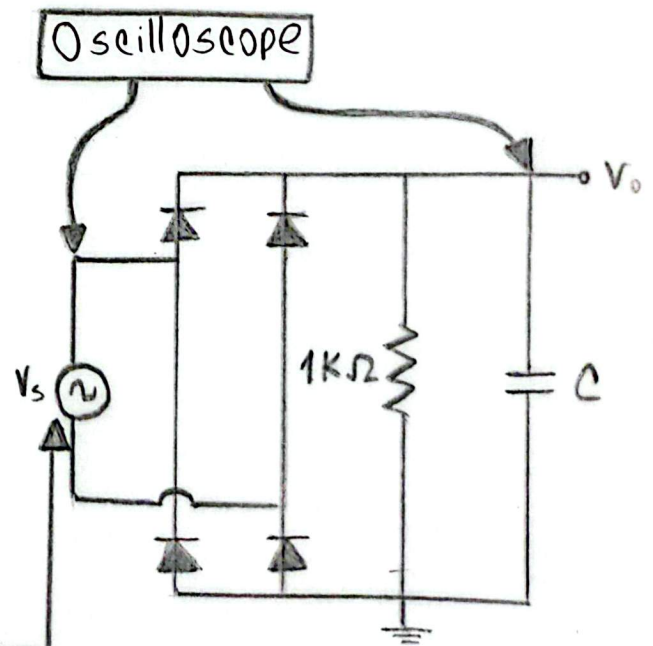
Required Apparatus:

- Diode (1N4007, 4 pieces)
- Resistor ($1\text{ k}\Omega$, 1 piece)
- Capacitor ($1\text{ }\mu\text{F}$, $10\text{ }\mu\text{F}$: 1 piece each)
- Center tapped transformer (1 piece)
- DC supply (0-15V)
- Signal generator
- Oscilloscope
- Project board
- Multimeter
- Connecting wires

Circuit Diagram:



(a) Half-Wave Rectifier



(b) Full-Wave Bridge Rectifier

Procedure:

- (i) The circuit was connected according to the circuit diagram without capacitor (Figure: 01)
- (ii) The input and output voltages in the oscilloscope were observed and recorded
- (iii) Repeated step (ii) after connecting a $1\text{ }\mu\text{F}$ capacitor and observed how the output waveform changes
- (iv) Repeated step (iii) but this time with a $10\text{ }\mu\text{F}$ capacitor
- (v) After that, the circuit was reassembled and then again connected according to the circuit diagram without capacitor (Figure: 02)
- (vi) The input and output voltages in the oscilloscope were observed and recorded
- (vii) Repeated step (vi) after connecting a $1\text{ }\mu\text{F}$ capacitor and observed how the output waveform changes
- (viii) Repeated step (vii) but this time with a $10\text{ }\mu\text{F}$ capacitor

Discussion and conclusion:

This experiment confirmed that the half-wave rectifier delivers only one half of the AC input, resulting in low DC output and high ripple, while the full-wave rectifier utilizes both halves and provides a higher, smoother DC level. Adding capacitors significantly reduces ripple, with the 10 μ F capacitor giving the most stable output. Overall, the results align with theory and clearly show the superior efficiency and output quality of full-wave and capacitor filtered rectifiers.

Precautions:

- All the connections were correctly given
- Oscilloscope were calibrated properly